

(DEEP) LEARNING TO TRADE
BY GUFLER, SANGIORGI, AND TARANTINO

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MOTIVATION

AI agents could take over markets

AI learns differently than people

QUESTIONS

How do AI traders affect markets?

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Price efficiency?

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Liquidity?

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Price efficiency?

Liquidity?

Competition?

THIS PAPER

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Part 1: Demand system for “representative investor”

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Qualitatively: Similar to rational BM

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Mkt eff and liq increase in wealth and comp

Part 3: AI experiments with calibrated residual demand

Qualitatively: Similar to rational BM

Quantitatively: Mkt eff and liq lower than in rational BM

COMMENTS

1. COMMENTS ON DEMAND SYSTEM

COMMENT 1.1: “REP INVESTOR”

Authors assume representative investor demand function of characteristics

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Suggestion: Model investor optimization and equity characteristics

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Matters: Key quantities authors analyze

Suggestion: Micro-founded model with roles for these quantities from get go

2. COMMENTS ON R.E. BENCHMARK

COMMENT 2.1: EXTREME BENCHMARK?

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Question: Conceptual or real-world benchmark?

Matters: Motivation suggests real world

Suggestion: Benchmarks with limited information or cognition

COMMENT 2.2: WHO ARE SPECULATORS?

Authors treat speculators as exogenous to demand system

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Suggestion: Model interaction explicitly

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Suggestion: Make definitions more axiomatic

3. COMMENTS ON EXPERIMENTS

COMMENT 3.1: AI OR UNCERTAINTY?

Authors model AI as different from BM in two ways:

- (i) Don't know DGP
- (ii) Reinforcement learning (non-Bayesian “clipping”)

But only (ii) defines AI

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E.g., Bayesian learning in BM

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Suggestion: BMs with one agent; strategic uncertainty under Bayesianity,...

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Suggestion: Consider optimal dynamic learning

COMMENT 3.4: ROLE OF REAL DATA

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Matters: Determines quantitative relevance

Suggestion: Motivate reason for real data or use simpler laboratory

CONCLUSION

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1. Demand system

2. Benchmark

3. AI

1.1 “Rep investor”

2.1 Extreme BM

3.1 AI vs. uncertainty

1.2 Well suited?

2.2 Speculators, who?

3.2 Strat. uncertainty

2.3 Definitions

3.3 Optimal learning

3.4 Real data